Version 7.4.8

PVsyst - Simulation report

Standalone system

Project: marwan_omar

Variant: New simulation variant
Standalone system with batteries
System power: 480 Wp
Dakar/Taklou - Senegal

Author

Project summary

Geographical Site
Dakar/Taklou
Senegal

Situation

Latitude	14.74 °N
Longitude	-17.48 °W
Altitude	20 m
Time zone	UTC-1

Project settings

Albedo	0.20

Weather data
Dakar/Yoff
MeteoNorm 7.2 station - Synthetic

System summary

Standalone system

Standalone system with batteries

PV Field Orientation
Fixed plane

Tilt/Azimuth	15 / 0 °

User's needs
Daily household consumers
Seasonal modulation

Average	2.3 kWh/Day

System information

PV Array

Nb. of modules	8 units
Pnom total	480 Wp

Battery pack

Technology	Lead-acid, sealed, Gel
Nb. of units	4 units
Voltage	24 V
Capacity	320 Ah

Results summary

Useful energy from solar	784.05 kWh/year	Specific production	1633 kWh/kWp/year	Perf. Ratio PR	74.89 %
Missing Energy	59.39 kWh/year	Available solar energy	810.52 kWh/year	Solar Fraction SF	92.96 %
Excess (unused)	10.99 kWh/year				

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General parameters

Standalone system **Standalone system with batteries**

PV Field Orientation

Orientation		**Sheds configuration**	**Models used**	
Fixed plane		No 3D scene defined	Transposition	Perez
Tilt/Azimuth	15 / 0 °		Diffuse	Perez, Meteonorm
			Circumsolar	separate

User's needs

Daily household consumers
Seasonal modulation

Average	2.3 kWh/Day

PV Array Characteristics

PV module

Manufacturer	Generic
Model	Poly 60 Wp 36 cells
(Original PVsyst database)	
Unit Nom. Power	60 Wp
Number of PV modules	8 units
Nominal (STC)	480 Wp
Modules	4 string x 2 In series
At operating cond. (50°C)	
Pmpp	432 Wp
U mpp	30 V
I mpp	14 A

Battery

Manufacturer	Generic
Model	Solar 12V / 160 Ah
Technology	Lead-acid, sealed, Gel
Nb. of units	2 in parallel x 2 in series
Discharging min. SOC	20.0 %
Stored energy	6.1 kWh
Battery Pack Characteristics	
Voltage	24 V
Nominal Capacity	320 Ah (C10)
Temperature	Fixed 20 °C

Controller

Universal controller	
Technology	MPPT converter
Temp coeff.	-5.0 mV/°C/Elem.
Converter	
Maxi and EURO efficiencies	97.0 / 95.0 %

Battery Management control

Threshold commands as	SOC calculation
Charging	SOC = 0.90 / 0.75
approx.	26.4 / 25.1 V
Discharging	SOC = 0.20 / 0.45
approx.	23.6 / 24.4 V

Total PV power

Nominal (STC)	0.480 kWp
Total	8 modules
Module area	4.5 m²
Cell area	3.7 m²

Array losses

Thermal Loss factor

Module temperature according to irradiance

Uc (const)	20.0 W/m²K
Uv (wind)	0.0 W/m²K/m/s

DC wiring losses

Global array res.	36 mΩ
Loss Fraction	1.5 % at STC

Serie Diode Loss

Voltage drop	0.7 V
Loss Fraction	2.1 % at STC

Module Quality Loss

Loss Fraction	2.5 %

Module mismatch losses

Loss Fraction	0.5 % at MPP

Strings Mismatch loss

Loss Fraction	0.1 %

IAM loss factor

ASHRAE Param.: IAM = 1 - bo (1/cosi -1)

bo Param.	0.05

Detailed User's needs

Daily household consumers, Seasonal modulation, average = 2.3 kWh/day

Summer (Jun-Aug)

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps (LED or fluo)	6	10/lamp	4.0	240
TV / PC / Mobile	1	75/app	3.0	225
Domestic appliances	1	200/app	3.0	600
Fridge / Deep-freeze	1		24	799
Dish- and Cloth-washer	1		1	1
Stand-by consumers			24.0	144
Total daily energy				2009

Autumn (Sep-Nov)

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps (LED or fluo)	6	10/lamp	5.0	300
TV / PC / Mobile	1	75/app	4.0	300
Domestic appliances	1	200/app	4.0	800
Fridge / Deep-freeze	1		24	799
Stand-by consumers			24.0	144
Total daily energy				2343

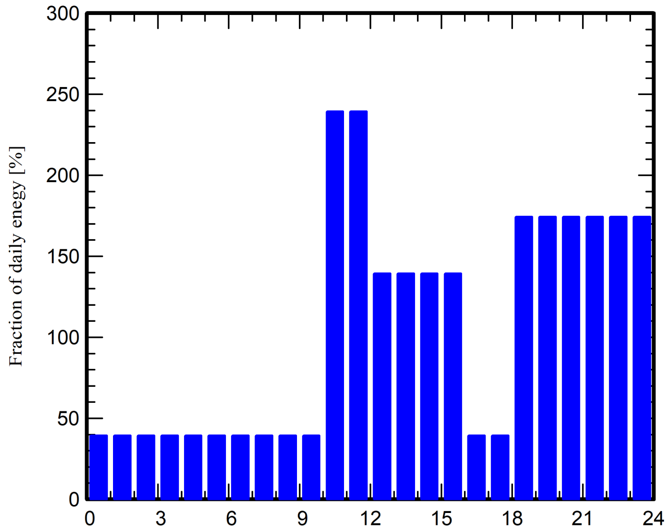
Winter (Dec-Feb)

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps (LED or fluo)	6	10/lamp	6.0	360
TV / PC / Mobile	1	75/app	6.0	450
Domestic appliances	1	200/app	4.0	800
Fridge / Deep-freeze	1		24	799
Stand-by consumers			24.0	144
Total daily energy				2553

Spring (Mar-May)

	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps (LED or fluo)	6	10/lamp	5.0	300
TV / PC / Mobile	1	75/app	4.0	300
Domestic appliances	1	200/app	4.0	800
Fridge / Deep-freeze	1		24	799
Stand-by consumers			24.0	144
Total daily energy				2343

Hourly distribution

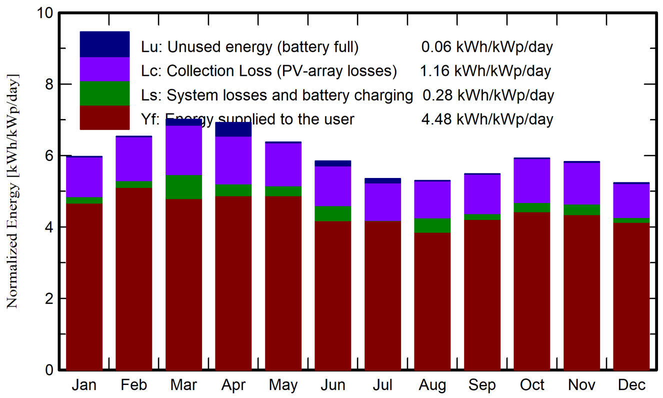


Main results

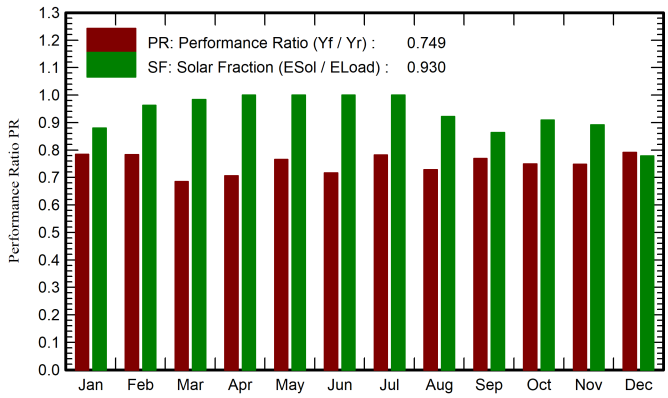
System Production

Useful energy from solar	784.05 kWh/year	Perf. Ratio PR	74.89 %
Available solar energy	810.52 kWh/year	Solar Fraction SF	92.96 %
Excess (unused)	10.99 kWh/year		
Loss of Load		**Battery aging (State of Wear)**	
Time Fraction	0.0 %	Cycles SOW	96.6 %
Missing Energy	59.39 kWh/year	Static SOW	90.0 %

Normalized productions (per installed kWp)



Performance Ratio PR



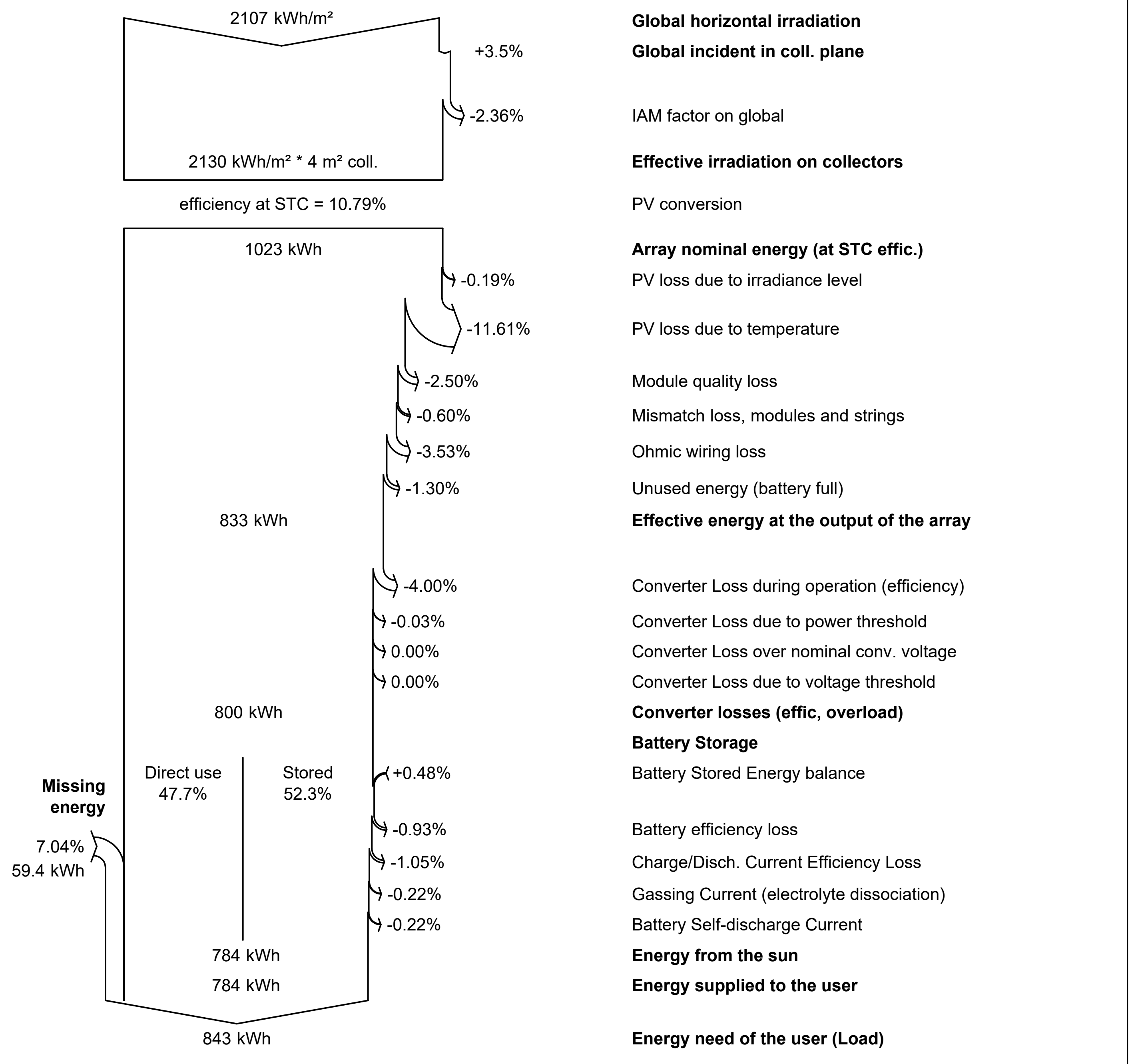
Balances and main results

	GlobHor kWh/m²	GlobEff kWh/m²	E_Avail kWh	EUnused kWh	E_Miss kWh	E_User kWh	E_Load kWh	SolFrac ratio
January	159.5	181.0	69.49	0.000	9.47	69.68	79.15	0.880
February	164.7	179.1	68.44	0.000	2.66	68.83	71.49	0.963
March	207.0	212.8	80.49	2.194	1.15	71.49	72.64	0.984
April	209.7	202.9	77.34	5.218	0.00	70.30	70.30	1.000
May	209.2	192.6	73.81	0.126	0.00	72.64	72.64	1.000
June	188.0	170.5	65.52	1.766	0.00	60.28	60.28	1.000
July	176.4	161.3	61.48	1.658	0.00	62.29	62.29	1.000
August	168.7	159.9	60.76	0.003	4.84	57.45	62.29	0.922
September	161.7	160.7	60.49	0.001	9.56	60.74	70.30	0.864
October	170.8	179.7	67.09	0.000	6.57	66.07	72.64	0.910
November	153.1	170.8	64.41	0.010	7.60	62.69	70.30	0.892
December	137.9	158.3	61.20	0.009	17.55	61.60	79.15	0.778
Year	2106.7	2129.6	810.52	10.986	59.39	784.05	843.44	0.930

Legends

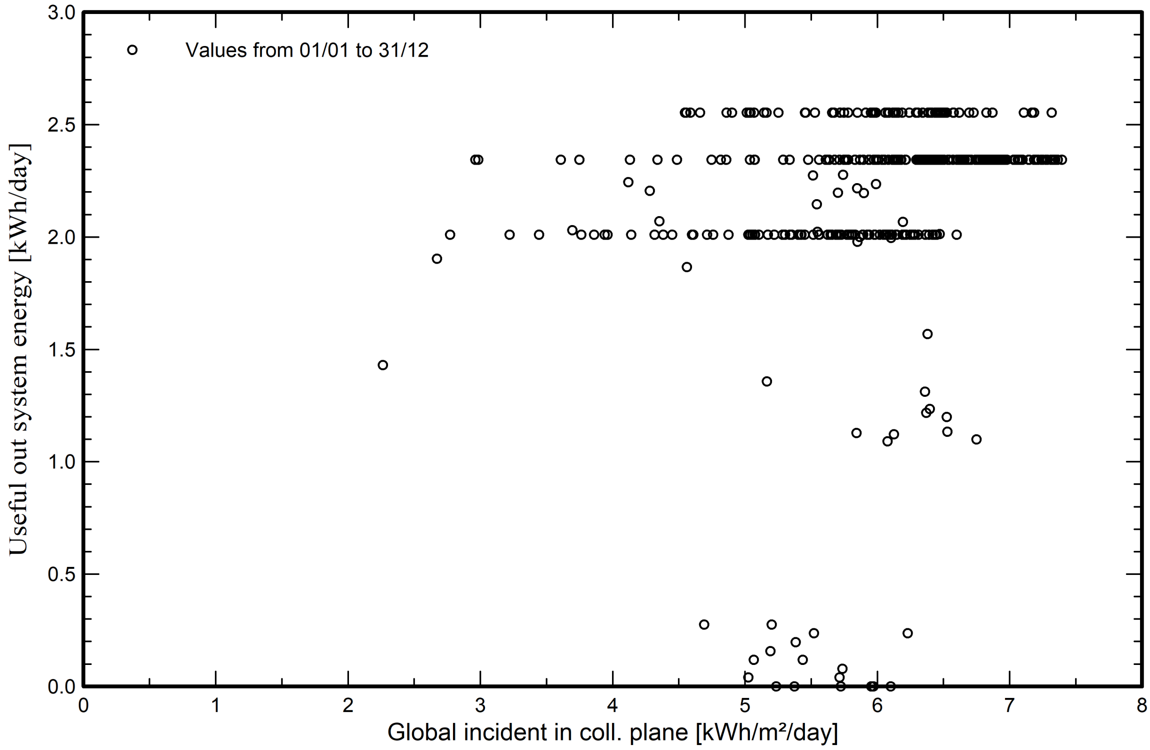
GlobHor	Global horizontal irradiation	E_User	Energy supplied to the user
GlobEff	Effective Global, corr. for IAM and shadings	E_Load	Energy need of the user (Load)
E_Avail	Available Solar Energy	SolFrac	Solar fraction (EUsed / ELoad)
EUnused	Unused energy (battery full)		
E_Miss	Missing energy		

Loss diagram



Predef. graphs

Daily Input/Output diagram



Cost of the system

Installation costs

Item	Quantity units	Cost EUR	Total EUR
		Total	0.00
		Depreciable asset	0.00

Operating costs

Item	Total
	EUR/year
Total (OPEX)	0.00

System summary

Total installation cost	0.00 EUR
Operating costs	0.00 EUR/year
Excess energy (battery full)	11.0 kWh/year
Used solar energy	784 kWh/year
Used energy cost	0.0000 EUR/kWh